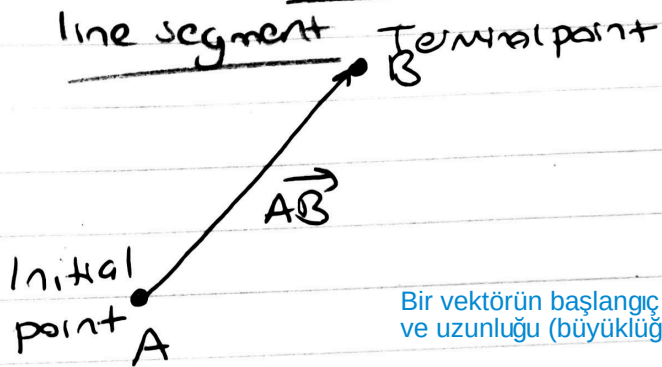


1.2.2 Vectors in $\mathbb{R}^3$ ($\mathbb{R}^3$ te vektörler)

A quantity such as force, displacement, or velocity is called a vector and is represented by a directed line segment.



Kuvvet, yer değiştirme veya hız gibi niceliklere vektör denir ve yönlü bir doğru parçası olarak gösterilir.

Bir vektörün başlangıç noktası A, bitiş noktası B ise bu vektör $\vec{AB}$ şeklinde gösterilir ve uzunluğu (büyüklüğü) $|\vec{AB}|$ ile ifade edilir.

İki vektör ancak ve ancak aynı uzunluğa ve aynı yöne sahipse birbirine eşittir.

Definitions: A vector is a directed line segment. The directed line segment $\vec{AB}$ has initial point A and terminal point B; its length is denoted by $|\vec{AB}|$. Two vectors are equal if they have the same length and direction.

Any vector $\vec{v} = \langle v_1, v_2, v_3 \rangle$ can be written as a linear combination of the standard unit vectors: $\vec{v} = v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k} = \langle v_1, v_2, v_3 \rangle$ component form

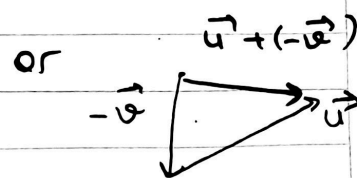
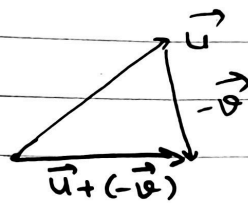
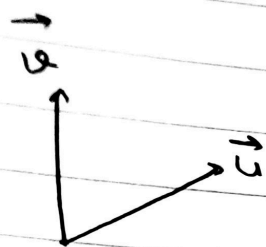
↖ bileşen form

Basic vectors : $\vec{i} = \langle 1, 0, 0 \rangle$

(unit vectors) $\vec{j} = \langle 0, 1, 0 \rangle$

birim vektör

$\vec{k} = \langle 0, 0, 1 \rangle$



Ex: $\vec{u} = -\vec{i} + 3\vec{j} + \vec{k}$, $\vec{v} = 4\vec{i} + 7\vec{j}$

(a) $2\vec{u} + 3\vec{v} = (2\vec{i} + 6\vec{j} + 2\vec{k}) + (12\vec{i} + 21\vec{j})$
 $= 14\vec{i} + 27\vec{j} + 2\vec{k}$

(b) $\vec{u} - \vec{v} = -5\vec{i} - 4\vec{j} + \vec{k}$

(c) $|\frac{1}{2}\vec{u}| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{1}{2}\sqrt{11}$
 $\left(\frac{1}{2}\vec{u} = -\frac{1}{2}\vec{i} + \frac{3}{2}\vec{j} + \frac{1}{2}\vec{k}\right)$

Properties of Vector Operations: Özellikler:

1) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

9) $(a+b)\vec{v} = a\vec{u} + b\vec{v}$

2) $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$

3) $\vec{u} + \vec{0} = \vec{u}$

4) $\vec{u} + (-\vec{u}) = \vec{0}$

5) $0\vec{u} = \vec{0}$

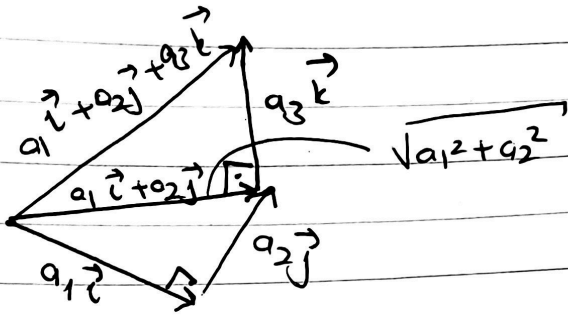
6) $1\vec{u} = \vec{u}$

7) $a(b\vec{u}) = (ab)\vec{u}$

8) $a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$

$$\vec{A} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$$

(*)



$$|\vec{A}| = \sqrt{a_1^2 + a_2^2 + a_3^2} \quad \text{length of magnitude}$$

uzunluk

k: sabit

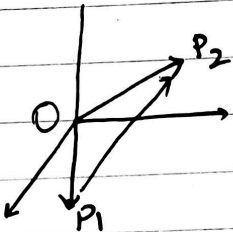
$$k \vec{A} = k a_1 \vec{i} + k a_2 \vec{j} + k a_3 \vec{k} \quad (k \text{ 'scales})$$

$$\vec{A} - \vec{B} = (a_1 - b_1) \vec{i} + (a_2 - b_2) \vec{j} + (a_3 - b_3) \vec{k}$$

$$(*) \quad \vec{0} = 0 \vec{i} + 0 \vec{j} + 0 \vec{k} \quad \text{zero vector} \quad \text{sıfır vektörü}$$

no direction, zero length
yönü yok

- İki Nokta Arasındaki Vektör: Başlangıç noktası $P_1(x_1, y_1, z_1)$ ve bitiş noktası $P_2(x_2, y_2, z_2)$ olan bir vektör $\vec{P_1 P_2} = \vec{OP_2} - \vec{OP_1} = (x_2 - x_1) \vec{i} + (y_2 - y_1) \vec{j} + (z_2 - z_1) \vec{k}$ şeklinde hesaplanır.



$\vec{OP_1}$ } position vectors
 $\vec{OP_2}$ } initial point at the origin

$$\begin{aligned} \vec{P_1 P_2} &= \vec{OP_2} - \vec{OP_1} \\ &= (x_2 \vec{i} + y_2 \vec{j} + z_2 \vec{k}) \\ &\quad - (x_1 \vec{i} + y_1 \vec{j} + z_1 \vec{k}) \\ &= (x_2 - x_1) \vec{i} + (y_2 - y_1) \vec{j} + (z_2 - z_1) \vec{k} \end{aligned}$$

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} \quad \text{unit vector (same direction with } \vec{v} \text{)}$$

$|\vec{u}| = 1$

- Birim Vektör: Bir $\vec{v}$ vektörüyle aynı yöndeki birim vektör $\vec{u} = \frac{\vec{v}}{|\vec{v}|}$ formülüyle hesaplanır ve uzunluğu daima 1'dir ($|\vec{u}| = 1$). Herhangi bir vektör, uzunluğu ile yönünün (birim vektörün) çarpımı olarak ifade edilebilir: $\vec{v} = |\vec{v}| \cdot \frac{\vec{v}}{|\vec{v}|}$.

$$\vec{v} = \underbrace{|\vec{v}|}_{\text{length}} \cdot \underbrace{\frac{\vec{v}}{|\vec{v}|}}_{\text{direction}}$$

Ex: Express $\vec{A} = \vec{i} - 2\vec{j} + 3\vec{k}$ as a product of its length and direction

$$|\vec{A}| = \sqrt{1^2 + (-2)^2 + 3^2} = \sqrt{14}$$

$$\vec{A} = |\vec{A}| \cdot \frac{\vec{A}}{|\vec{A}|} = \underbrace{\sqrt{14}}_{\text{length}} \left(\underbrace{\frac{1}{\sqrt{14}}\vec{i} - \frac{2}{\sqrt{14}}\vec{j} + \frac{3}{\sqrt{14}}\vec{k}}_{\text{direction}} \right)$$

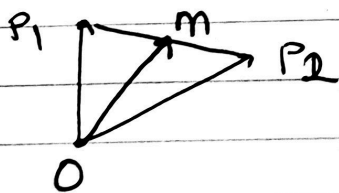
Ex: Find a unit vector $\vec{u}$ in the direction of the vector from $P_1(1, 0, 1)$ to $P_2(3, 2, 0)$.

$$\vec{P_1P_2} = (3-1)\vec{i} + (2-0)\vec{j} + (0-1)\vec{k} = 2\vec{i} + 2\vec{j} - \vec{k}$$

$$|\vec{P_1P_2}| = \sqrt{2^2 + 2^2 + (-1)^2} = 3$$

$$\vec{u} = \frac{\vec{P_1P_2}}{|\vec{P_1P_2}|} = \frac{2}{3}\vec{i} + \frac{2}{3}\vec{j} - \frac{1}{3}\vec{k}$$

Midpoints: Orta Nokta Arasındaki Pozisyon Vektörü:



$$\vec{Om} = \vec{OP_1} + \frac{1}{2}(\vec{P_1P_2})$$

$$= \vec{OP_1} + \frac{1}{2}(\vec{OP_2} - \vec{OP_1})$$

$$= \frac{1}{2}(\vec{OP_1} + \vec{OP_2})$$

$$= \frac{x_1 + x_2}{2}\vec{i} + \frac{y_1 + y_2}{2}\vec{j} + \frac{z_1 + z_2}{2}\vec{k}$$

Ex: The midpoint of the segment joining $P_1(3, -2, 0)$ and $P_2(7, 4, 4)$ is

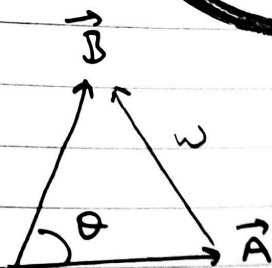
$$\left(\frac{3+7}{2}, \frac{-2+4}{2}, \frac{0+4}{2} \right) = (5, 1, 2)$$

12.3

Dot (Skalar) Products: İç Çarpım

Angle between two vectors iki vektör arasındaki açı

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta \quad (\text{number!})$$



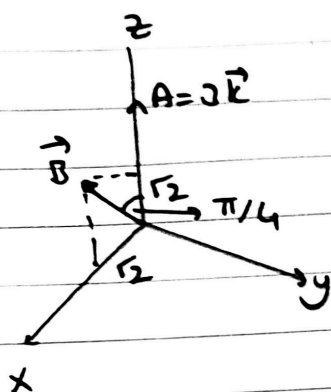
$$|W|^2 = |A|^2 + |B|^2 - 2|A||B|\cos \theta$$

$$2|A||B|\cos \theta = |A|^2 + |B|^2 - |W|^2$$

$$|W| = |A - B|$$

Ex:

$$\vec{A} = 3\vec{e}, \vec{B} = \sqrt{2}\vec{i} + \sqrt{2}\vec{j}, \vec{A} \cdot \vec{B}$$



$$\vec{A} \cdot \vec{B} = 3 \cdot 2 \cdot \cos \frac{\pi}{4} = 3\sqrt{2}$$

(*) The dot product is positive if the angle between vectors is acute, dar açı

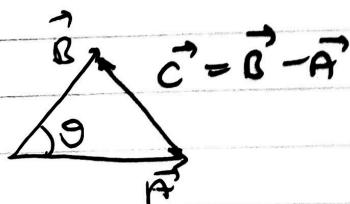
" " " negative if " " vectors is obtuse, geniş açı

$$(*) \vec{A} \cdot \vec{A} = |\vec{A}| |\vec{A}| \cos 0 = |\vec{A}|^2$$

$$\Rightarrow |\vec{A}| = \sqrt{\vec{A} \cdot \vec{A}}$$

Calculation:

Use law of cosines



$$|C|^2 = |A|^2 + |B|^2 - 2|A||B|\cos \theta$$

$$\vec{A} \cdot \vec{B}$$

$$\vec{A} = \langle a_1, a_2, a_3 \rangle$$

$$\vec{B} = \langle b_1, b_2, b_3 \rangle$$

$$\vec{A} \cdot \vec{B} = \frac{|\vec{A}|^2 + |\vec{B}|^2 - |\vec{C}|^2}{2}$$

$$\vec{A} \cdot \vec{B} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{|\vec{A}| \cdot |\vec{B}|} \right)$$

Ex: $\vec{A} = \langle 1, -2, -2 \rangle, \vec{B} = \langle 6, 3, 2 \rangle$

$$\vec{A} \cdot \vec{B} = 6 - 6 - 4 = -4$$

$$|\vec{A}| = \sqrt{1+4+4} = 3, \quad |\vec{B}| = \sqrt{36+9+4} = 7$$

$$\theta = \cos^{-1} \left(\frac{-4}{3 \cdot 7} \right) = \cos^{-1} \left(-\frac{4}{21} \right)$$

Laws: Kurallar:

1) $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ (commutativity)

$$a_1 b_1 + a_2 b_2 + a_3 b_3 = b_1 a_1 + b_2 a_2 + b_3 a_3$$

2) c : any number

$$(c \vec{A}) \cdot \vec{B} = \vec{A} \cdot (c \vec{B}) = c (\vec{A} \cdot \vec{B})$$

3) $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$

$$(\vec{A} + \vec{B}) \cdot \vec{C} = \vec{A} \cdot \vec{C} + \vec{B} \cdot \vec{C}$$

4) $\vec{A} \cdot \vec{A} = |\vec{A}|^2$

5) $\vec{0} \cdot \vec{A} = 0$

Perpendicular Vectors: ^{→ orthogonal}

Dik (Ortogonal Vektörler)

$$\Rightarrow) \vec{A} \perp \vec{B} \Rightarrow \cos \frac{\pi}{2} = 0 \Rightarrow \vec{A} \cdot \vec{B} = 0 \\ = (|\vec{A}| |\vec{B}| \cos \frac{\pi}{2})$$

$$\Leftarrow) \vec{A} \cdot \vec{B} = \underbrace{|\vec{A}|}_{\neq 0} \underbrace{|\vec{B}|}_{\neq 0} \cos \theta = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

$$\Rightarrow \vec{A} \perp \vec{B}$$

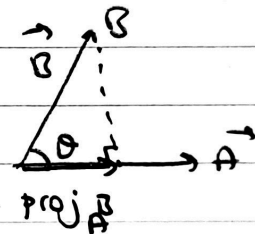
$$\boxed{\vec{A} \perp \vec{B} \Leftrightarrow \vec{A} \cdot \vec{B} = 0}$$

Ex: $\vec{A} = 3\vec{i} - 2\vec{j} + \vec{k}$, $\vec{B} = 2\vec{j} + 4\vec{k}$

$$\vec{A} \cdot \vec{B} = 3 \cdot 0 + (-2) \cdot 2 + 1 \cdot 4 = 0 \Rightarrow \vec{A} \perp \vec{B}$$

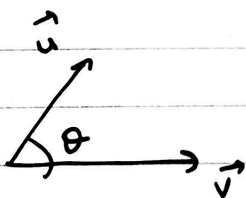
Projection Vectors: ^{iz düşüm vektörleri}

$$\text{Length of } \text{proj}_{\vec{A}} \vec{B} = |\vec{B}| \cos \theta$$



$$\text{proj}_{\vec{V}} \vec{U} = \underbrace{(|\vec{U}| \cos \theta)}_{\text{length}} \underbrace{\frac{\vec{V}}{|\vec{V}|}}_{\text{direction}}$$

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$$



$$= \underbrace{\left(\frac{\vec{U} \cdot \vec{V}}{|\vec{V}|} \right)}_{\text{Scalar comp.} \atop \text{skaler bileşen}} \frac{\vec{V}}{|\vec{V}|} = \left(\frac{\vec{U} \cdot \vec{V}}{|\vec{V}|} \right) \frac{\vec{V}}{|\vec{V}|} = \left(\frac{\vec{U} \cdot \vec{V}}{|\vec{V}|^2} \right) \vec{V}$$

Vector projection of $\vec{u}$ onto $\vec{v}$

$$\text{proj}_{\vec{v}} \vec{u} : \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v}$$

u vektörünün

v vektörü üzerine izdüşüm vektörü şu formülle hesaplanır:

Ex: $\vec{u} = 6\vec{i} + 3\vec{j} + 2\vec{k}$, $\vec{v} = \vec{i} - 2\vec{j} - 2\vec{k}$

$$\text{proj}_{\vec{v}} \vec{u} = \left(\frac{6 - 6 - 4}{1 + 4 + 4} \right) (\vec{i} - 2\vec{j} - 2\vec{k})$$

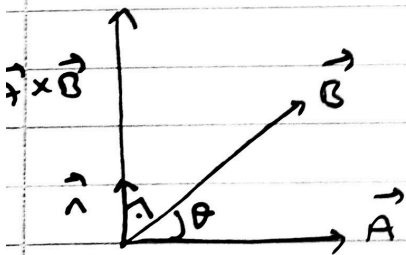
$$= -\frac{4}{9} \vec{i} + \frac{8}{9} \vec{j} + \frac{8}{9} \vec{k}$$

$$\text{Scalar component} = \vec{u} \cdot \frac{\vec{v}}{|\vec{v}|} = 2 - 2 - \frac{4}{3} = -\frac{4}{3}$$

*12.4 Cross Product: Vektörel Çarpım

$\vec{A}, \vec{B}$: nonzero vectors (not parallel!)

$\vec{n}$: unit normal vector birim normal vektör



$$\vec{A} \times \vec{B} = (|\vec{A}| |\vec{B}| \sin \theta) \vec{n}$$

$\vec{A} \times \vec{B}$ is a scalar multiple of $\vec{n}$

$$\vec{A} \times \vec{B} \perp \vec{A}, \vec{A} \times \vec{B} \perp \vec{B}$$

Parallel Vectors:

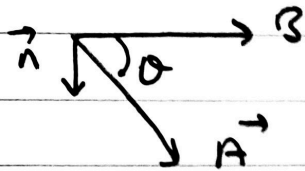
$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A} \times \vec{B} = \vec{0}$$

$$(\Rightarrow) \theta = 0, \pi \Rightarrow \sin \theta = \sin \pi = 0 \Rightarrow \vec{A} \times \vec{B} = \vec{0}$$

$$\Leftarrow \vec{A} \times \vec{B} = \vec{0} \Rightarrow \sin \theta = 0 \Rightarrow \theta = 0, \pi$$

$$\Rightarrow \vec{A} \parallel \vec{B}$$

NOTE: $\vec{B} \times \vec{A} = -(\vec{A} \times \vec{B})$ (not commutative)



$$\vec{i} \times \vec{j} = -(\vec{j} \times \vec{i}) = \vec{k}$$

$$\vec{j} \times \vec{k} = -(\vec{k} \times \vec{j}) = \vec{i}$$

$$\vec{k} \times \vec{i} = -(\vec{i} \times \vec{k}) = \vec{j}$$

$$\vec{i} \times \vec{i} = \vec{j} \times \vec{j} = \vec{k} \times \vec{k} = \vec{0}$$

(by def.)

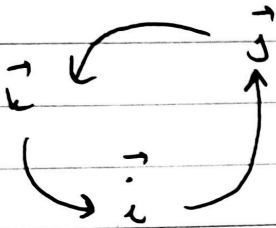
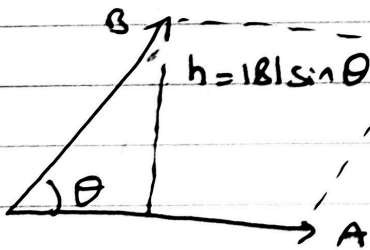


diagram for
recalling these products.

$$(*) |\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \underbrace{|\sin \theta|}_1 = |\vec{A}| |\vec{B}| \sin \theta$$



$$\begin{aligned} \text{Area} &= \text{base} \cdot \text{Height} \\ &= |\vec{A}| \cdot |\vec{B}| \sin \theta \\ &= |\vec{A} \times \vec{B}| \end{aligned}$$

(Area of parallelogram)

paralel kenarın uzunluğu

$$A \times B = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

• Bu determinant, kofaktör açılımı yöntemiyle şu şekilde çözülür:

$$A \times B = \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$A \times B = (a_2 b_3 - a_3 b_2) \vec{i} - (a_1 b_3 - a_3 b_1) \vec{j} + (a_1 b_2 - a_2 b_1) \vec{k}$$